Multi-Party Distributed Point Functions, Revisited

Elaine Shi, Tianyao Gu, Xuanye Zheng

Yue Yang, Yiping Liu, Yucheng Fu

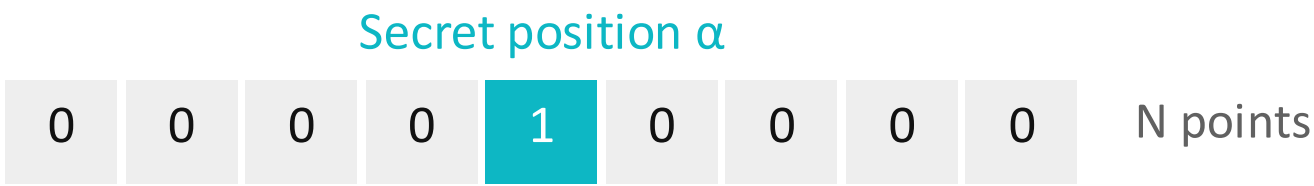
Randomized author order

Carnegie Mellon University

University of Virginia

Point functions

$$f(x) = 1 \text{ if } x = \alpha$$
$$f(x) = 0 \text{ otherwise}$$



The nonzero value is assumed to be 1 for simplicity.

Distributed point functions (DPFs)

Secret sharing of a point function

Gen(α): gives a share k_i to party i

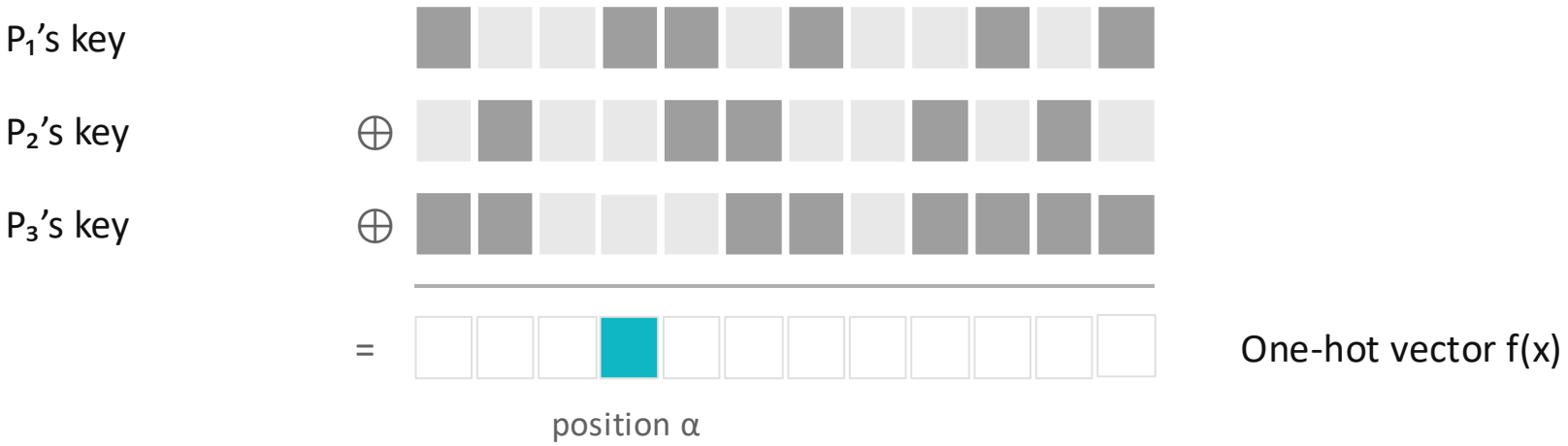
Eval(k_i, x): party i uses their share to obtain shares y_i of f(x)

$$y_1 \oplus \cdots \oplus y_m = f(x).$$

Security: any t shares hide the target point α.

The naive DPF

Each party holds a random length-N vector that XOR to the point function.



Perfect secrecy against all but one parties.

The challenge: shares much smaller than N.

Applications

- Private information retrieval [GI14] [BGI16]
- Anonymous messaging [CGBM15]
- Function secret sharing (DPF is a special case) [BCG+21] [BGI15]
- Multi-party computation [BGI19] [BGH+26]
- Private set intersection [GGM24]
- Pseudorandom correlation generators [BCG+19] [BCG+20]
- Distributed ORAM [DS17] [BKKO20]

Two-party vs. multi-party DPFs

Two-party

$\mathrm{polylog}(N)$

share size from OWF

[GI14] [BGI16]

Multi-party

$\approx \sqrt{N}$

share size from OWF

[BGI15] [GWW25]

Our contributions

Asymptotic improvements

Honest-majority DPF

- Removes exponential dependence on num parties

Dishonest-majority DPF

- Reduce dependence on num parties from m^3 to $\sqrt{m}$

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Security amplification paradigm

$(1-\delta)$–secure DPF

⬇

$(1-\mathrm{negl}(\lambda))$–secure DPF

e.g. [BGIK22b] [GWW25]

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e.g. [BGIK22b] [GWW25]

Problem: gap in [BGIK22b]'s proof.

Our work: a new, complete proof.

Our contributions

Asymptotic improvements

Honest-majority DPF **This talk**

- **Removes exponential dependence on num parties**

Dishonest-majority DPF

- Reduce dependence on num parties from m^3 to $\sqrt{m}$

Security amplification paradigm

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$(1-\mathrm{negl}(\lambda))$–secure DPF

e.g. [BGIK22b] [GWW25]

Problem: gap in [BGIK22b]'s proof.

Our work: a new, complete proof.

Honest-majority DPF [BKO22]

Assume m = 2t + 1 for simplicity.

[BKO22] achieves share size

$$O_\lambda\left(\sqrt{\binom{m}{t}} \cdot N^{\frac{1+\varepsilon}{4}}\right)$$

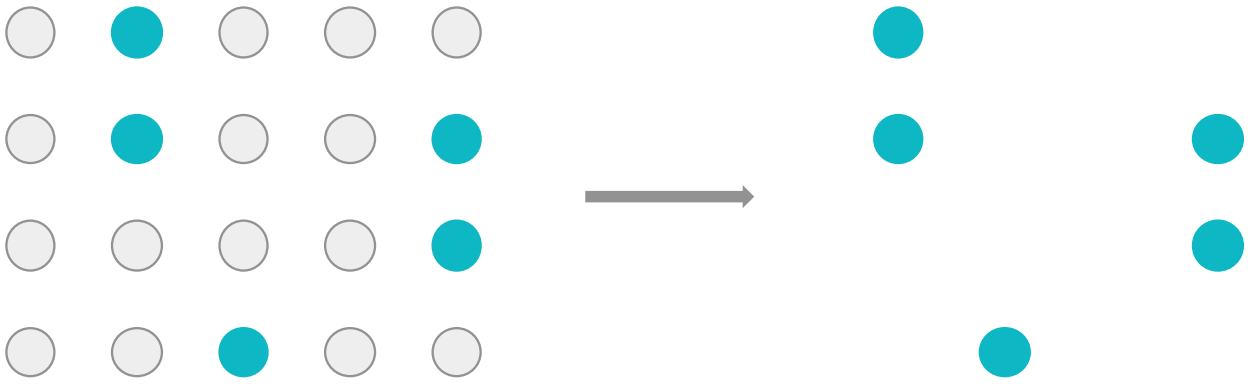
$$= 2^{\Theta(m)}$$

exponential in m

To reduce the cost, we use committee selection.

Honest-majority DPF (ours)

Committee selection



m parties

k committee members

Everyone else outputs 0.

Honest-majority DPF (ours)

Committee selection

k	Share size $2^{\theta(k)}$	Pr[committee is honest majority]
$\text{polylog}(\lambda)$ Traditional parameter choice	$N^{\frac{1+\varepsilon}{4}} \cdot 2^{\text{polylog}(\lambda)}$	$1 - \text{negl}(\lambda)$

Honest-majority DPF (ours)

Committee selection

k	Share size $2^{\theta(k)}$	Pr[committee is honest majority]
$\text{polylog}(\lambda)$ Traditional parameter choice	$N^{\frac{1+\varepsilon}{4}} \cdot 2^{\text{polylog}(\lambda)}$	$1 - \text{negl}(\lambda)$
$\log \log(\lambda)$ Our choice	$N^{\frac{1+\varepsilon}{4}} \cdot \text{polylog}(\lambda)$	$1 - \frac{1}{\text{polylog}(\lambda)}$

1/polylog leakage can be plugged by security amplification.

DPF security amplification

Weakly secure DPF

1/polylog
security loss



Small efficiency loss

Fully secure DPF

Negligible
security loss

- Proposed by [BGIK22b] with incomplete proof
- Our paper completes the proof and restores the prior results, e.g., [GWW25]

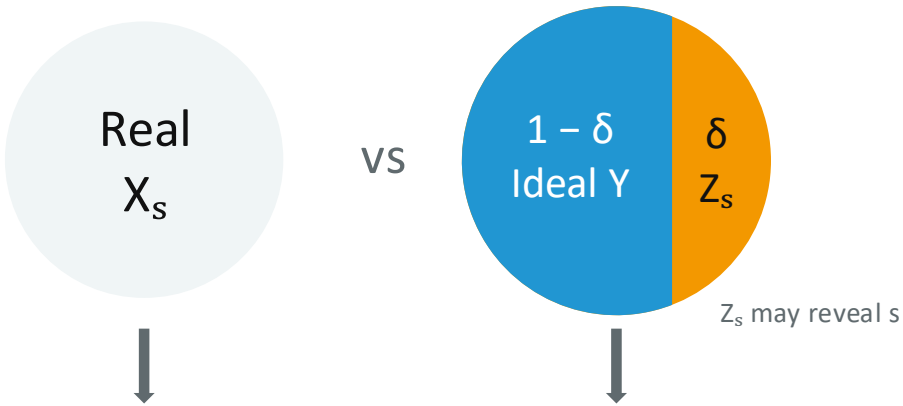
Stepping stone: strong δ-indistinguishability

Weak δ-indistinguishability



Efficient adversary has advantage ≤ δ

Strong δ-indistinguishability



Efficient adversary has negligible advantage

Summary

Scheme	Per-party share size
Honest majority: m > 2t, D = ⌊m/t⌋, D′ = ⌊(1 − ε)m/t⌋	
[BKO22] + [GWW25]	$O_\lambda(N^{1/(2D')} \cdot \binom{m}{t}^3)$
Our work	$\tilde{O}_\lambda(N^{(1+\varepsilon)/(2D')})$
Dishonest majority: t = m − 1	
[BGI15]	$O_\lambda(2^{m/2} \cdot N^{1/2})$
[GWW25]	$\tilde{O}_\lambda(N^{1/2+\varepsilon} \cdot m^3)$
Our work	$\tilde{O}_\lambda(N^{1/2+\varepsilon} \cdot \sqrt{m})$

N: domain size m: number of parties t: corrupt parties ε: any small constant

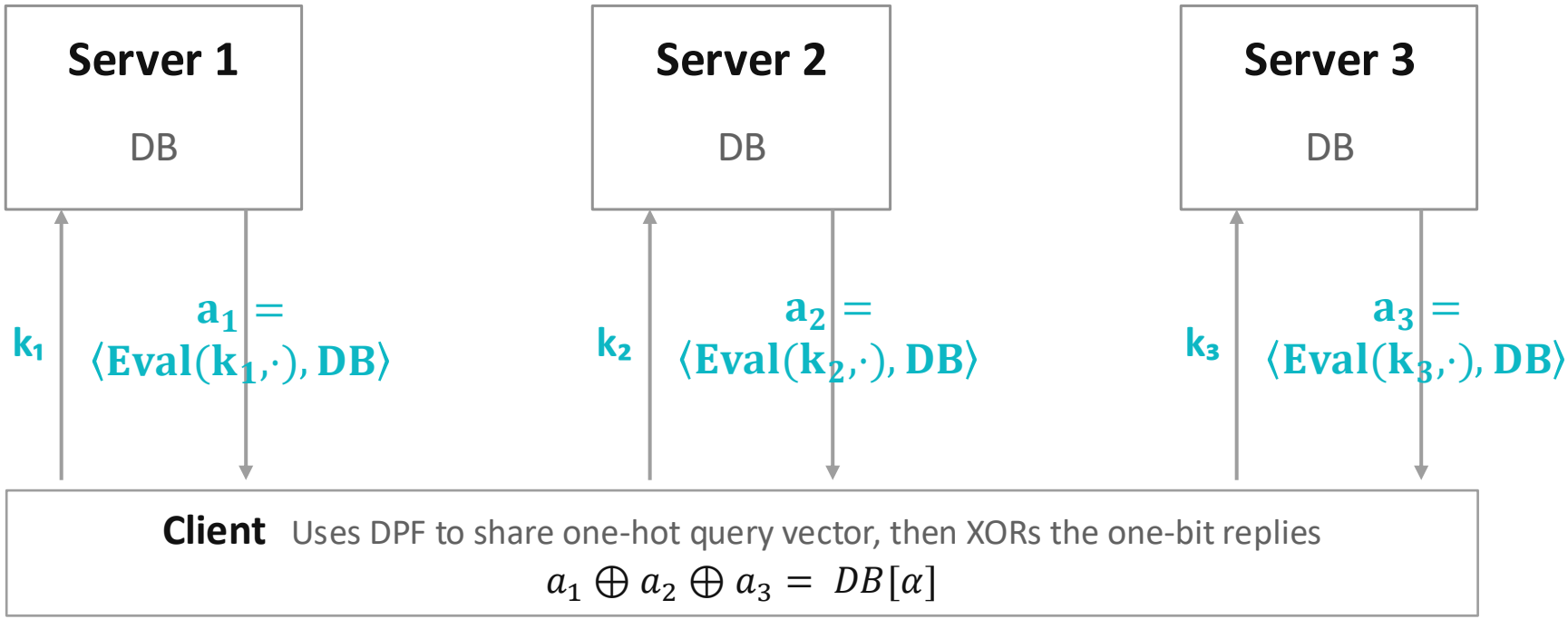
All schemes assume only one-way functions.

Refer to our paper for security amplification details.

Thank you!

Application: Private Information Retrieval (PIR)

PIR goal: client obtains DB[α] without revealing α.



Short DPF keys give short private queries.

References (1/5): Main body

[GI14] Niv Gilboa and Yuval Ishai. Distributed Point Functions and Their Applications. *EUROCRYPT 2014,* pp. 640–658.

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[BCG+21] Elette Boyle, Nishanth Chandran, Niv Gilboa, Divya Gupta, Yuval Ishai, Nishant Kumar, and Mayank Rathee. Function Secret Sharing for Mixed-Mode and Fixed-Point Secure Computation. *EUROCRYPT 2021,* pp. 871–900.

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[GRS22] Gayathri Garimella, Mike Rosulek, and Jaspal Singh. Structure-Aware Private Set Intersection, with Applications to Fuzzy Matching. *CRYPTO 2022,* pp. 323–352.

[GGM24] Gayathri Garimella, Benjamin Goff, and Peihan Miao. Computation Efficient Structure-Aware PSI from Incremental Function Secret Sharing. *CRYPTO 2024,* pp. 309–345.

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[BCG+20] Elette Boyle, Geoffroy Couteau, Niv Gilboa, Yuval Ishai, Lisa Kohl, and Peter Scholl. Efficient Pseudorandom Correlation Generators from Ring-LPN. *CRYPTO 2020,* pp. 387–416.

[DS17] Jack Doerner and abhi shelat. Scaling ORAM for Secure Computation. *ACM CCS 2017,* pp. 523–535.

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